Miniproject Report

on

SMART MONITORING SYSTEM FOR SOLITARY-LIVING AND ELDERLY INDIVIDUALS

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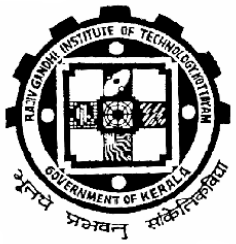
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Certificate

This is to certify that the Project Report entitled **Smart Monitoring System for Solitary-Living and Elderly Individuals** is a bonafide work carried out by **ADITHYAN P.U.(KTE23EC071)**, **ADHILA FATHIMA T.A.(KTE23EC010)**, **NANDANALAKSHMI P.H. (KTE23EC052)**, and **NANDANA NAIR (KTE23EC053)** during 2025–26, in partial fulfillment for the award of the **B.Tech. Degree in Electronics and Communication Engineering** of A.P.J. Abdul Kalam Technological University, Rajiv Gandhi Institute of Technology, Kottayam.

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ABSTRACT

The project addresses the growing safety concerns of individuals living alone, particularly elderly people, who may not receive timely assistance during emergencies. Situations such as prolonged inactivity or inability to respond can go unnoticed due to the lack of continuous supervision, increasing the risk of serious health or safety issues.

To address this problem, a smart non-wearable monitoring system is proposed that observes user activity during predefined time intervals set by the user. The system identifies abnormal inactivity and provides a mechanism for user interaction, ensuring that emergency situations are recognised effectively. Upon detecting such conditions, alerts are automatically sent to predefined contacts to enable timely assistance.

The proposed system is designed to be simple, reliable, and user-friendly, while maintaining user comfort and privacy. It offers a cost-effective solution for enhancing the safety and well-being of individuals living independently and can be further extended for real-world home monitoring applications.

Contents

ACKNOWLEDGEMENT **i**

ABSTRACT **ii**

List of Figures **v**

1 INTRODUCTION **1**

2 LITERATURE REVIEW **3**

3 SYSTEM DESCRIPTION **5**

3.1 Block Diagram 6

3.2 Working Principle 7

3.3 Functional Overview 7

3.3.1 System Initialization 7

3.3.2 Time-Based Monitoring Activation 8

3.3.3 Activity Detection using Sensor Fusion 8

3.3.4 Decision Making and Condition Evaluation 8

3.3.5 Alert Generation and Communication 8

3.3.6 Manual Override and User Interaction 9

3.3.7 Continuous Operation Loop 9

4 SOFTWARES USED **10**

4.1 Arduino IDE 10

4.2 Embedded C/C++ 10

4.3 Proteus 11

4.4 Serial Monitor 11

5 HARDWARE DESCRIPTION **13**

5.1 ESP32 Microcontroller 13

5.2 PIR Sensor 14

5.3 Ultrasonic Sensor 14

5.4 RTC Module 15

5.5 GSM Module 15

5.6 Keypad 16

5.7 SOS Button 17

5.8 LCD Display 17

5.9 EEPROM 18

6 RESULTS **19**

6.1 Motion Detection Performance 19

6.2 Time-Based Monitoring Efficiency 20

6.3 User Interaction and Input 20

6.4 Mobile Application and Settings 20

6.5 Overall System Performance 21

7 CONCLUSION **23**

7.1 Future Scope 23

REFERENCES **24**

List of Figures

3.1 System Block Diagram 6

4.1 Circuit Diagram 11

5.1 ESP32 Microcontroller 13
5.2 PIR Sensor 14
5.3 Ultrasonic Sensor 14
5.4 RTC Module 15
5.5 GSM Module 15
5.6 Keypad 16
5.7 SOS Button 17
5.8 LCD Display 17
5.9 EEPROM 18

6.1 System workflow 19
6.2 SoloCare Mobile Application 21
6.3 Smart Monitoring Device 22

Chapter 1

INTRODUCTION

In today's fast-paced world, the number of individuals living alone—particularly elderly people and working professionals—has increased significantly. While independent living promotes freedom and self-reliance, it also introduces serious safety concerns, especially during unforeseen situations where immediate assistance may not be available. Even a short period of unresponsiveness during critical times of the day can indicate potential health risks, and the absence of timely intervention can lead to severe consequences.

Existing monitoring solutions largely depend on wearable devices, which are often inconvenient, uncomfortable, or easily neglected by users. This reduces their reliability in real-world scenarios. There is a growing need for a smarter, more practical approach that ensures user safety without requiring constant user involvement.

This project presents a non-wearable smart monitoring system that focuses on verifying user activity during scheduled time periods. By combining motion sensing, real-time processing, and emergency alert mechanisms, the system is capable of identifying abnormal inactivity and triggering timely alerts. With added features such as manual SOS activation and keypad-based contact entry, the system offers a user-friendly, accessible, and dependable solution aimed at improving safety, independence, and peace of mind for individuals living alone.

Motivation

Individuals living alone, particularly elderly people, require a reliable safety monitoring system. In emergency situations, they may be unable to call for help due to physical limitations or unconsciousness. Caregivers and family members cannot provide constant supervision, creating a need for an automated monitoring solution. Moreover, wearable devices often face issues related to user compliance and comfort. A non-wearable, low-cost system that can monitor activity and provide timely alerts can significantly improve safety and reduce response time during emergencies

Scope of the Project

The scope of this project encompasses the design, development, and testing of a smart, non-wearable monitoring system aimed at enhancing the safety of individuals living alone, particularly elderly users. It involves the detection of human activity during predefined time intervals using a Passive Infrared (PIR) sensor, combined with real-time monitoring and decision-making implemented through the ESP32 microcontroller. The system integrates a Real-Time Clock (RTC) module to ensure accurate time-based monitoring and evaluation of user responsiveness within scheduled periods.

The system incorporates a GSM communication module to transmit emergency alerts in the form of calls or messages to pre-registered contacts when abnormal inactivity or manual SOS activation is detected. Additionally, a keypad interface is integrated to enable users to input and update emergency contact details directly, improving system accessibility and independence. A 16×2 LCD display provides real-time system status and alerts.

Through the integration of sensing, processing, and communication technologies, the project aims to deliver a reliable, cost-effective, and user-friendly monitoring solution that ensures timely assistance and enhances safety in home environments.

Chapter 2

LITERATURE REVIEW

Non-intrusive monitoring systems have gained significant attention in recent years. In [1], the authors present a system for recognizing daily human activities using environmental sensors deployed in indoor environments. The study emphasises the use of low-power, non-wearable sensing techniques that ensure user comfort and privacy. The system demonstrates reliable activity recognition over time, making it suitable for elderly monitoring applications. This approach highlights the effectiveness of non-wearable sensors, which aligns with the proposed system that utilizes PIR sensors for motion detection.

In [2], a comprehensive review of non-wearable sensor-based monitoring systems for elderly care is presented. The study explores various environmental sensing technologies integrated with IoT and edge intelligence frameworks for detecting abnormal behavior. It highlights how non-wearable sensors can continuously monitor user activity without causing discomfort, making them suitable for long-term usage. The paper also discusses the advantages and limitations of such systems, particularly in terms of accuracy and feature extraction. This work supports the proposed system, which uses environmental sensors to monitor user activity and detect inactivity over defined intervals.

In [3], the authors investigate human movement detection using Passive Infrared (PIR) sensors. The study analyzes analog output signals from PIR sensors to track movement patterns, including direction and speed. It demonstrates that PIR sensors can effectively detect human presence and support activity recognition without requiring wearable devices. However, the study also identifies limitations

in detecting stationary individuals and achieving high precision. These findings are directly relevant to the proposed system, where PIR sensors are used for motion detection, complemented by additional sensing techniques to improve reliability.

Privacy-preserving monitoring systems using non-visual sensors are explored in [4]. The authors propose a system based on PIR sensors and other low-cost technologies to detect human presence and inactivity without using cameras. The study highlights that PIR sensors are effective for indoor monitoring while maintaining user privacy. This directly supports the use of PIR sensors in the proposed system for safe and privacy-aware monitoring.

In [5], a GSM-based patient monitoring system is introduced, focusing on real-time communication of emergency conditions. The system integrates sensors with communication modules to send alerts via calls and messages to caregivers. The study emphasises the importance of reliable communication during emergencies. This aligns with the proposed system, which uses GSM technology to notify emergency contacts when abnormal inactivity or SOS activation is detected.

Chapter 3

SYSTEM DESCRIPTION

The proposed system is a smart, non-wearable monitoring solution designed to ensure the safety of individuals living alone by verifying their activity during predefined time intervals. The system is built around the ESP32 microcontroller, which acts as the central processing unit and coordinates the operation of all connected components.

The system primarily uses a Passive Infrared (PIR) sensor to detect human motion within the monitored area. During scheduled monitoring periods set using the Real-Time Clock (RTC) module, the system checks for user activity. If motion is detected, the system confirms that the user is active and functioning normally. However, if no movement is detected within a specified duration, the system identifies a potential inactivity condition.

To minimise false alerts, a response window is provided during which the user can acknowledge their presence or cancel the alert. The system also includes a physical SOS button that allows users to manually trigger an emergency alert when needed.

A 4x4 matrix keypad is integrated into the system to allow users to input and update emergency contact numbers directly. This feature enhances usability by removing the need for external configuration through a mobile device.

Once an emergency condition is confirmed, either automatically due to inactivity or manually through the SOS feature, the GSM module sends alert messages or initiates calls to pre-registered contacts. The 16×2 LCD display provides real-time feedback on system status, monitoring activity, and alert conditions.

Overall, the system combines sensing, processing, user input, and communication modules to provide a reliable, cost-effective, and user-friendly solution for monitoring individuals in a home environment.

3.1 Block Diagram

The block diagram of the system is shown in Figure 3.1. It illustrates the main functional modules, including sensing, processing, user interface, storage, and communication units.

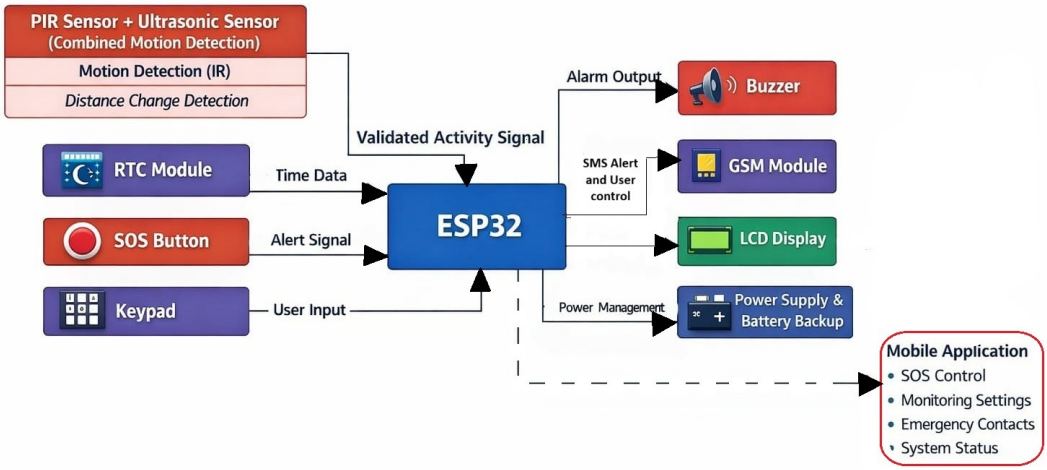


Figure 3.1: System Block Diagram

The PIR and ultrasonic sensors detect movement and presence within the monitored area and send signals to the ESP32 controller. The controller processes these inputs and evaluates user activity based on predefined time intervals. The keypad allows users to enter emergency contact numbers, which are stored in EEPROM for reliable access. If inactivity is detected, the system triggers an alert through the communication module. Mobile phone integration enables configuration and monitoring of system parameters.

3.2 Working Principle

The system operates by monitoring user activity within predefined time intervals using a combination of PIR and ultrasonic sensors. The PIR sensor detects motion based on infrared radiation, while the ultrasonic sensor measures distance variations, and their combined output improves detection accuracy. These signals are processed by the ESP32 microcontroller along with time data from the RTC module. If no valid activity is detected during the scheduled period, the system identifies it as a potential abnormal condition. In response, it triggers alerts through the GSM module and buzzer. Additional components such as the SOS button, keypad, and LCD display enhance user interaction and system functionality.

3.3 Functional Overview

The functional overview of the proposed system is divided into several subsections to provide a clear understanding of its operation. Each subsection describes a specific stage of the system, including initialisation, time-based monitoring, activity detection using sensor fusion, decision-making, alert generation, and user interaction. This structured approach helps in explaining the overall workflow of the system in a systematic and concise manner.

3.3.1 System Initialization

When the system is powered on, the ESP32 microcontroller initialises all connected modules, including the PIR sensor, ultrasonic sensor, RTC module, GSM module, keypad, buzzer, and LCD display. The system ensures proper communication between components and sets initial conditions for operation. The RTC module provides the current time, which is used to determine the scheduled monitoring intervals.

3.3.2 Time-Based Monitoring Activation

The system operates based on predefined monitoring periods rather than continuous observation. Using time data from the RTC module, the ESP32 checks whether the current time falls within the scheduled monitoring window. If the system is outside the defined time interval, it remains in an idle state, conserving power and avoiding unnecessary processing. Once the scheduled time begins, the monitoring process is activated.

3.3.3 Activity Detection using Sensor Fusion

During the monitoring phase, the system detects user activity using a combination of PIR and ultrasonic sensors. The PIR sensor identifies motion based on infrared radiation emitted by the human body, while the ultrasonic sensor measures changes in distance within the environment. The ESP32 processes inputs from both sensors and combines them to generate a validated activity signal. This sensor fusion approach significantly reduces false detections and improves the reliability of the system.

3.3.4 Decision Making and Condition Evaluation

The ESP32 continuously evaluates whether valid activity is detected within the defined time interval. If sufficient activity is observed, the system continues normal monitoring. However, if no activity is detected for a specific duration within the scheduled period, the system interprets it as a potential abnormal condition and prepares to trigger an alert.

3.3.5 Alert Generation and Communication

In case of abnormal inactivity or emergency detection, the system activates the GSM module to send alert messages or make calls to pre-stored emergency contacts. Simultaneously, a buzzer is triggered to provide a local audible alert. This ensures both remote and nearby notification of the situation.

3.3.6 Manual Override and User Interaction

The system includes an SOS button that allows the user to manually trigger an emergency alert at any time, independent of sensor inputs. Additionally, a keypad interface enables users to input or update emergency contact details. The LCD display provides real-time system status, including monitoring state, alerts, and user prompts, improving overall usability.

3.3.7 Continuous Operation Loop

After completing each monitoring cycle, the system updates the display and continues operation in a loop. It repeatedly checks time conditions, reads sensor inputs, and evaluates system status to ensure continuous and reliable monitoring during scheduled periods.

Chapter 4

SOFTWARES USED

The development of the monitoring system involves the use of various software tools for programming, simulation, and system testing. These tools provide a platform for implementing embedded logic, interfacing hardware components, and verifying system performance. Each software plays a specific role in ensuring the efficient design, debugging, and deployment of the system.

4.1 Arduino IDE

Arduino Integrated Development Environment (IDE) is an open-source software platform used for writing, compiling, and uploading code to microcontrollers. It supports multiple programming languages, primarily C and C++, and provides a wide range of built-in libraries for hardware interfacing. In this project, Arduino IDE is used to develop and upload code to the ESP32 microcontroller. It enables integration of various modules such as PIR sensor, ultrasonic sensor, RTC module, GSM module, keypad, buzzer, and LCD display. The IDE also provides a serial monitor feature, which is used for debugging and monitoring real-time system behaviour.

4.2 Embedded C/C++

Embedded C/C++ is a programming language widely used to develop firmware for microcontroller-based systems. It provides efficient control over hardware re-

sources and supports real-time execution of tasks. In this project, Embedded C/C++ is used to implement the system logic, including sensor data acquisition, signal processing, decision-making algorithms, and communication control. It ensures smooth coordination between different hardware components and enables reliable system performance.

4.3 Proteus

Proteus is a simulation software used for designing and testing electronic circuits in a virtual environment. It allows users to verify circuit connections and component behaviour before physical implementation. In this project, Proteus can be used to simulate the circuit design and validate the interfacing of different components, such as sensors and a microcontroller. This helps in identifying potential errors and improving system reliability before hardware development .

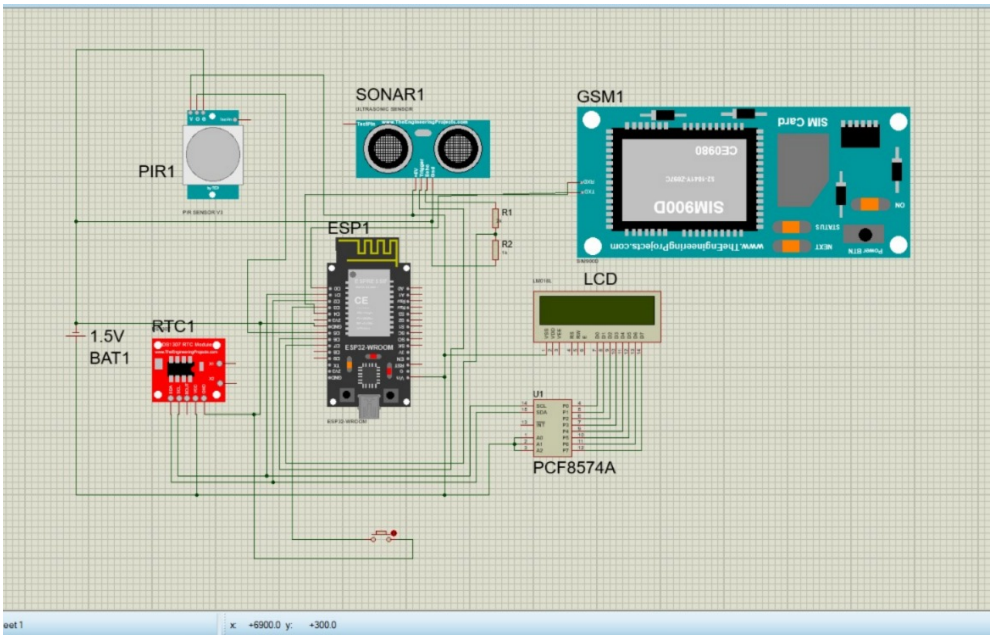


Figure 4.1: Circuit Diagram

4.4 Serial Monitor

The Serial Monitor is a tool integrated within the Arduino IDE that enables real-time communication between the microcontroller and the user. It is mainly used

for debugging and testing purposes. In this project, the Serial Monitor is used to observe sensor readings, system status, and decision outputs during execution. It helps in troubleshooting errors, validating system logic, and ensuring proper functioning of the entire system.

Chapter 5

HARDWARE DESCRIPTION

The system is built using a combination of sensors, control units, communication modules, and user interface components. Each hardware component plays a crucial role in ensuring accurate monitoring, decision-making, and alert generation. The integration of these components enables the system to function efficiently within scheduled monitoring periods while providing both automatic and manual emergency responses.

5.1 ESP32 Microcontroller



Figure 5.1: ESP32 Microcontroller

The ESP32 is a low-power microcontroller with built-in Wi-Fi and Bluetooth capabilities, widely used in embedded and IoT applications. It supports multiple GPIO pins and enables real-time data processing. In this project, the ESP32 serves as the central control unit, processing sensor inputs, managing time-based

operations, and controlling communication and alert mechanisms. It is chosen over other microcontrollers due to its higher processing capability, integrated connectivity features, and ability to handle multiple peripherals efficiently.

5.2 PIR Sensor

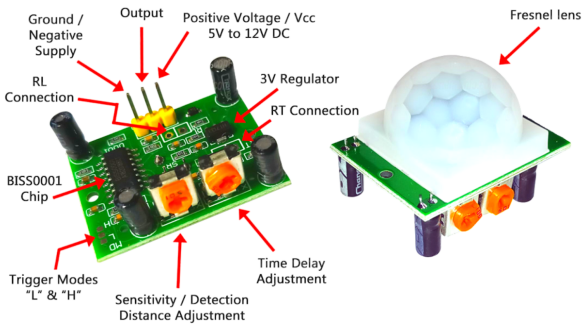


Figure 5.2: PIR Sensor

The Passive Infrared (PIR) sensor detects motion based on infrared radiation emitted by the human body. It is widely used in motion detection systems due to its simplicity and low power consumption. In this project, the PIR sensor is used to detect human movement within the monitored area and provides a digital signal to the ESP32 indicating motion presence. It is selected because of its low cost, energy efficiency, and reliable detection of human presence.

5.3 Ultrasonic Sensor



Figure 5.3: Ultrasonic Sensor

The ultrasonic sensor measures distance by transmitting ultrasonic waves and calculating the time taken for the echo to return. It is commonly used for proximity

and object detection. In this project, it works alongside the PIR sensor to detect changes in distance, helping confirm actual movement and improving detection accuracy through sensor fusion. It is chosen to reduce false detections and provide additional validation compared to using only a PIR sensor.

5.4 RTC Module



Figure 5.4: RTC Module

The Real-Time Clock (RTC) module keeps track of current time and date, even during power interruptions. It ensures accurate timekeeping for time-based applications. In this project, the RTC module is used to control scheduled monitoring by providing time data to the ESP32 for activating or deactivating the system. It is selected for its accuracy, reliability, and ability to support time-based automation.

5.5 GSM Module



Figure 5.5: GSM Module

The GSM module enables wireless communication using cellular networks, allowing the system to send SMS messages or make calls. In this project, it is used to send emergency alerts to predefined contacts when abnormal conditions or manual SOS activation occur. It is chosen due to its wide network coverage and ability to function independently of internet connectivity.

5.6 Keypad



Figure 5.6: Keypad

The keypad is an input device used to enter numerical or alphanumeric data into the system. It provides a simple interface for user interaction. In this project, the keypad allows users to manually input or update emergency contact details without modifying the program. It is selected for its ease of use, flexibility, and ability to enhance system accessibility.

5.7 SOS Button



Figure 5.7: SOS Button

The SOS button is a push-button switch used to manually trigger an emergency alert. It provides an immediate way to signal distress. In this project, pressing the SOS button bypasses normal monitoring conditions and directly activates the alert system. It is chosen for its simplicity, reliability, and quick response capability in emergency situations.

5.8 LCD Display

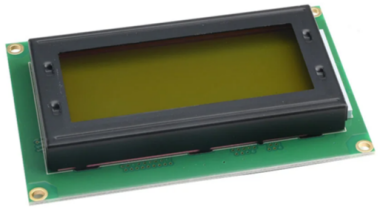


Figure 5.8: LCD Display

The LCD (Liquid Crystal Display) is used to display text-based information to the user. It enhances interaction by providing visual feedback. In this project, the LCDs system status, monitoring conditions, alert messages, and user prompts. It is chosen for its clarity, low power consumption, and ability to improve user experience.

5.9 EEPROM



Figure 5.9: EEPROM

EEPROM is a type of non-volatile memory used to store small amounts of data that must be retained even when the system is powered off. It allows data to be written, erased, and rewritten electrically. In this project, EEPROM is used to store user-defined emergency contact details entered through the keypad. This ensures that the data remains (retained) even after power loss, eliminating the need to re-enter information each time the system restarts. It is chosen due to its non-volatile nature, reliability, and suitability for storing small but critical configuration data compared to volatile memory like RAM.

Chapter 6

RESULTS

The system was successfully designed and implemented to monitor user activity within scheduled time intervals and generate alerts during abnormal conditions. The integration of PIR and ultrasonic sensors, along with time-based control using the RTC module, enabled accurate and reliable system performance. The system was tested under various conditions to evaluate its functionality, responsiveness, and accuracy.

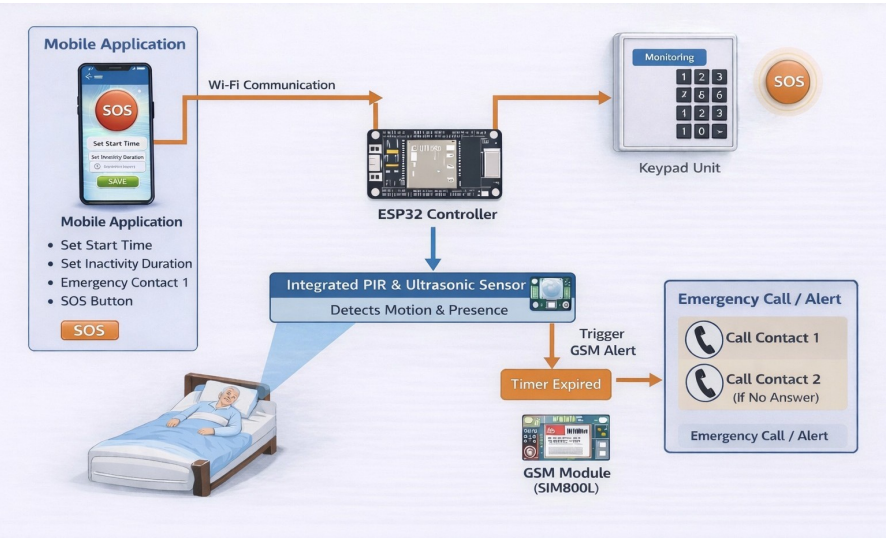


Figure 6.1: System workflow

6.1 Motion Detection Performance

The combination of PIR and ultrasonic sensors provided improved motion detection accuracy compared to using a single sensor. The PIR sensor effectively

detected human presence, while the ultrasonic sensor validated movement through distance variation. This sensor fusion approach significantly reduced false detections caused by environmental factors, ensuring reliable activity monitoring during scheduled periods.

6.2 Time-Based Monitoring Efficiency

The RTC module successfully controlled the activation and deactivation of the monitoring system based on predefined time intervals. The system remained inactive outside the scheduled period, thereby reducing unnecessary operation and improving efficiency. During active periods, the system continuously monitored activity as intended.

The GSM module effectively transmitted alert messages to pre-stored emergency contacts when no activity was detected within the defined duration or when the SOS button was pressed. The buzzer also generated immediate local alerts, ensuring both remote and nearby notification. The alert system demonstrated consistent performance with minimal delay.

6.3 User Interaction and Input

The keypad interface allowed users to enter and update emergency contact details easily. The integration of EEPROM ensured that the entered data was retained even after power interruptions. The LCD display provided real-time feedback, including system status and alert messages, enhancing overall usability.

6.4 Mobile Application and Settings

The mobile application interface provided additional flexibility and control over the system. Users were able to configure the monitoring parameters, including setting the time duration after which activity is evaluated during scheduled periods. This allowed customization based on individual user routines.

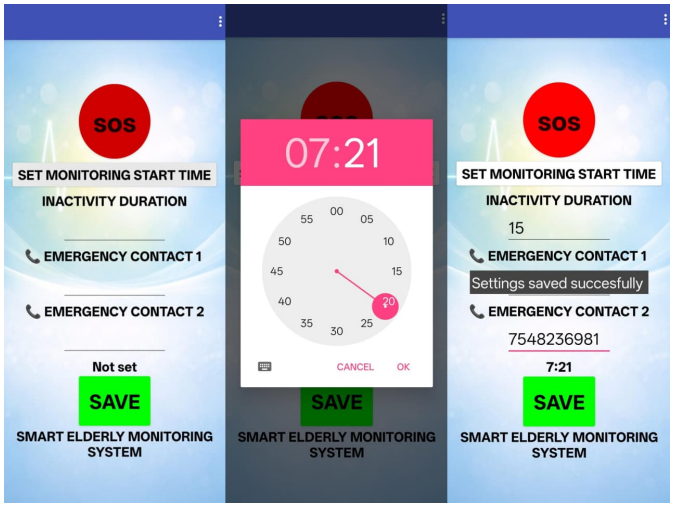


Figure 6.2: SoloCare Mobile Application

The application also enabled users to add and manage emergency contact details, ensuring that alerts are sent to the correct recipients. An in-app SOS button was included, allowing users to manually trigger emergency alerts remotely. The integration of these features improved the overall usability and adaptability of the system, making it more convenient for real-world usage.

6.5 Overall System Performance

The system operated reliably under different test scenarios, including normal activity, inactivity during scheduled periods, and manual emergency activation. The coordination between hardware components and software logic ensured smooth operation and accurate decision-making. The results demonstrate that the system is effective in providing a simple and reliable monitoring and alert solution.



Figure 6.3: Smart Monitoring Device

Chapter 7

CONCLUSION

The system successfully demonstrates a smart and efficient approach to monitoring user activity within predefined time intervals and generating alerts during abnormal conditions. By integrating PIR and ultrasonic sensors, the system achieves improved motion detection accuracy through sensor fusion, reducing false detections and enhancing reliability. The use of the RTC module enables time-based operation, ensuring that monitoring is performed only during scheduled periods, thereby optimising system efficiency.

The incorporation of a GSM module ensures timely communication with emergency contacts, while the buzzer provides immediate local alerts. Additional features, such as the keypad for manual data entry and EEPROM for data storage, enhance system usability and flexibility. The SOS button further strengthens the system by allowing manual emergency triggering.

Overall, the system provides a cost-effective, reliable, and user-friendly solution for activity monitoring and emergency alerting. The successful implementation and testing of the system highlight its potential for real-world applications in personal safety and assisted monitoring scenarios.

7.1 Future Scope

The system can be further enhanced by integrating IoT technology to enable real-time remote monitoring and alert notifications through mobile or cloud platforms. Additional sensors or wearable devices can be incorporated to improve activity

tracking and provide more comprehensive monitoring. Machine learning techniques can also be applied to analyse user behaviour patterns and reduce false alerts. Furthermore, features such as GPS-based location tracking and improved power efficiency can be implemented to enhance system reliability and portability. These advancements can make the system more intelligent, adaptive, and suitable for real-world applications.

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